

MSBA Ops Multi-Agent Dispatch Report

1 message

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19 May 2026 at 19:43

To: [redacted-email]

SeeWeeS Medical Logistics Dispatch Decision Memo

1. Executive Decision Summary

Apply a **25% travel-time / departure lead-time buffer** to the **Tier 1 C1_I95_NJ_BOS corridor** over the current **24–48 hour** planning horizon, based on a **route weather risk score of 2** and waypoint-level heavy rain and high wind flags. Maintain standard Tier 2 posture for **C2_NJ_PHL** with no additional corridor-specific buffer beyond existing policy.

Continue to **exclude all DQ-01 rows with missing unique_item_id** from dispatch optimization while counting and logging them, accepting a current exclusion rate of **11.1% on C1** and **6.7% on C2**. No additional human escalation is required at this time; leadership focus should be on **tight monitoring of Tier 1 SLA margin and DQ-01 trends** during the 2x volume step-up.

2. Decision Actions

- **Apply a 25% buffer to C1_I95_NJ_BOS:** Implement a 25% travel-time or departure lead-time buffer for all Tier 1 C1 shipments in the 24–48 hour window, consistent with the **weather_risk score 2** policy.
- **Keep DQ-01 rows out of optimization:** Exclude the **5 DQ-01 units** (including **2 on C1** and **1 on C2** in the planning window) from dispatch routing while maintaining full logging with `reason_code = excluded_missing_unique_item_id`.
- **Normalize all valid shipments to canonical_item_id:** Rely on the existing reconciliation outcome of **95 exact_match**, **26 alias_match**, and **3 legacy_id_map** rows, and keep these mappings active for all planning and KPI calculations.
- **Operate C1 and C2 at the higher volume baseline:** Treat **15 valid units/day** (vs. historical **7.83/day**) as the current planning assumption when assessing corridor-level SLA and capacity KPIs.

3. SLA Watch Items

- **Tier 1 C1_I95_NJ_BOS SLA exposure:** Corridor is Tier 1 with **16 valid planning-window units** under a **6-hour SLA exposure** and an applied **25% weather buffer**; SLA risk flags should be interpreted in that buffered context.
- **C1 exclusion visibility risk:** C1 has **2 excluded units out of 18 total** in the planning window (**11.1% exclusion rate**), all with `reason_code = excluded_missing_unique_item_id` (DQ-01), reducing traceability for a subset of Tier 1 flow.
- **Tier 2 C2_NJ_PHL SLA visibility:** C2 carries **14 valid** and **1 excluded** unit in the planning window (**6.7% exclusion rate**), representing lower but present data-quality impact on Tier 2 SLA visibility.
- **Volume step-up effect:** Planning-window average daily valid units are **15.0** vs. historical **7.83**; SLA monitoring should account for this **~2x volume increase** across both corridors when reviewing **ON_TRACK**, **AT_RISK**, and **BREACH_RISK** patterns.

4. Weather Route Snapshot

Waypoint	City	Risk Score (0–3)	Max Precip (mm/day)	Max Wind Gust (km/h)	Min Temp (°C)	Heavy Rain Risk	High Wind Risk	Freezing Risk
W1	Newark	0	0.0	43.2	20.1	false	false	false
W2	Bronx	2	21.2	48.6	20.1	true	true	false

Waypoint	City	Risk Score (0–3)	Max Precip (mm/day)	Max Wind Gust (km/h)	Min Temp (°C)	Heavy Rain Risk	High Wind Risk	Freezing Risk
W3	New Haven	1	4.3	63.0	16.4	false	true	false
W4	Providence	1	0.0	49.7	18.3	false	true	false
W5	Boston	1	0.7	50.8	17.6	false	true	false

5. Corridor Performance Snapshot

Corridor	Default SLA Tier	Planning-Window Valid Units	Planning-Window Excluded Units	Planning-Window Total Units	Excluded Rate
C1_I95_NJ_BOS	Tier 1	16	2	18	11.1%
C2_NJ_PHL	Tier 2	14	1	15	6.7%

6. Rule-Grounded Rationale

- **Weather buffer policy:** The `route_risk_score_0_3 = 2` for C1_I95_NJ_BOS, with `heavy_rain_risk` and `high_wind_risk` at W2 (Bronx), maps to a **25% travel-time / departure lead-time buffer** per the stated weather buffer policy for Tier 1 corridors; escalation is only specified at score 3.
- **DQ-01 and exception handling:** All **5 excluded units** (2 on C1, 1 on C2 in the planning window, plus 2 in history) have `reason_code = excluded_missing_unique_item_id` and `issue_codes = [DQ-01]`, aligning with DQ-01 and D1 precedence to remove rows with missing `unique_item_id` from dispatch calculation while counting and logging them.
- **Canonical reconciliation:** The data-quality corrections of **95 exact_match**, **26 alias_match**, and **3 legacy_id_map** rows follow D3–D5 and D2 by normalizing to `canonical_item_id` for planning and KPIs, with no use of substitution (D7) or `generated_identifier` in the current dataset.

7. Expected KPI Impact

- **Tier 1 SLA margin:** The 25% buffer on C1_I95_NJ_BOS at `weather_risk` score 2 is expected to be visible in corridor-level SLA risk flags, supporting ON_TRACK classifications for Tier 1 shipments under current conditions.
- **DQ-01 containment:** Maintaining exclusion of the **5 DQ-01 units** keeps rows with missing `unique_item_id` out of dispatch optimization while preserving them in data-quality and SLA KPI denominators.
- **Traceability via canonical keys:** Using `canonical_item_id` for all **124 valid units** (with 29 corrected via alias/legacy mappings) supports consistent corridor-level KPIs and reconciliation across the historical and planning windows.

8. Monitoring Triggers

- **Weather escalation:** Re-review dispatch and buffer settings if the C1_I95_NJ_BOS `route_risk_score_0_3` increases to **3** or if additional waypoints reach risk score ≥ 2 with `heavy_rain_risk` or `high_wind_risk`.
- **DQ-01 trend:** Trigger a data-quality review if corridor exclusion rates rise materially above the current **11.1% on C1** or **6.7% on C2**, or if total excluded units significantly exceed the current **5 of 129 rows (3.9%)**.
- **SLA risk pattern:** Reassess corridor buffers and scheduling if deterministic SLA risk flags show multiple C1 shipments classified as AT_RISK or BREACH_RISK despite the 25% buffer.
- **Volume pressure:** Re-examine corridor KPIs and capacity assumptions if planning-window valid units per day move materially above the current **15.0** average while `weather_risk` for C1 remains at or above score 2.
- **Reconciliation load:** Initiate an upstream item-master review if the share of `alias_match` and `legacy_id_map` corrections grows significantly beyond the current **29 corrected units** out of **124 valid**, signaling drift in source item naming or IDs.